A CIRCLE, CONICS AND A LEMNISCATE MEET IN THE BAR ... EHH I MEAN CRITICAL LINE

TacoYaci 

tacoyaci.github.io

Abstract. Take the Riemann zeta function, written as the alternating series divided by $1-2^{1-s}$, let s walk up a vertical line, and watch each term $t_n(s)$ of this sum draw a curve. The first two terms draw circles, and their sum draws a circle around the origin exactly when s walks along the critical line. Behind this is a small mirror, since inversion in the circle of radius $1/\sqrt{2}$ swaps the pictures for the real parts a and $1-a$. The later terms refuse to draw conics, but the fourth one has a surprise for us. On the critical line it draws a perfect lemniscate of Bernoulli, i.e. an infinity sign.

1. Introduction

In this note I want to give some additions to the nice exposition [Ha]. It starts from the alternating series for the Riemann zeta function

$$\zeta(s) = \frac{1}{1-2^{1-s}} \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^s},$$

which converges for $\mathrm{Re}(s) > 0$, and the formula holds whenever $2^{1-s} \neq 1$. Moving the prefactor into the series, [Ha] looks at the single terms

$$t_n(s) = \frac{(-1)^{n-1}}{n^s\,(1-2^{1-s})} \qquad (n \geq 1)\,. \tag{1}$$

Write $s = a + ib$ with real a and b. For fixed a, the number $t_n(a+ib)$ moves through the complex plane as b runs through $\mathbb{R}$, and I call the set of these values the trace of t_n on the line $\mathrm{Re}(s) = a$. In [Ha], it is shown that t_1 and t_2 trace circles and that the circle traced by $t_1 + t_2$ is centered at the origin only on the critical line $a = 1/2$. On the critical line, grouping t_m with t_{2m} for odd m gives more circles around the origin, with radii $1/\sqrt{2m}$. It is also said there that the traces of all terms form "descending families of conics".

The first theorem is the main observation of [Ha] (see Figure 1).

Theorem 🍊**.** *Let $a \neq 1$. The trace of $t_1 + t_2$ on the line $\mathrm{Re}(s) = a$ is the circle with*

$$\textit{center}\ \ \frac{1-2^{1-2a}}{1-4^{1-a}} \qquad \textit{and radius}\ \ \frac{2^{-a}}{|1-4^{1-a}|}\,.$$

It is centered at 0 if and only if $a = 1/2$, and then its radius is $1/\sqrt{2}$.

The reason for the special role of $a = 1/2$ is a mirror. Inversion in the circle of radius $1/\sqrt{2}$ around 0, i.e. the map $w \mapsto 1/(2\overline{w})$, sends the value of $t_1 + t_2$ at $a + ib$ to its value at $(1-a) + ib$ (Proposition 2.1). On the critical line both values agree, so the trace is the mirror itself. The radius printed in [Ha, (11)] is different for $a \neq 0$, see Section 6.

The second theorem shows that the terms with $n \geq 3$ do not trace conics. Here a conic means the set of zeros of a nonzero real polynomial of degree at most 2 in the real and the imaginary part, so circles and lines count as conics.

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Theorem 🦄**.** *Let $n \geq 3$ and $a \neq 1$. Then the trace of t_n on the line $\mathrm{Re}(s) = a$ does not lie on a conic. If n is not a power of 2, the trace is dense in the annulus*

$$\frac{n^{-a}}{1+2^{1-a}} \leq |w| \leq \frac{n^{-a}}{|1-2^{1-a}|}\,.$$

This corrects the statement about conics in [Ha, Section 2] (see Section 6). The term t_4 still hides a nice classical curve.

Theorem ♾️**.** *The trace of t_4 on the critical line is the lemniscate of Bernoulli*

$$\left((x-\tfrac{1}{2})^2+y^2\right)^2 = \tfrac{1}{2}\left((x-\tfrac{1}{2})^2-y^2\right), \qquad w = x+iy\,.$$

For $a \neq 1/2$, the trace of t_4 on the line $\mathrm{Re}(s) = a$ is not a lemniscate of Bernoulli.

In fact $1/(t_4 - 4^{-a})$ lies on a conic for every $a \neq 1$, and on the critical line this conic is a rectangular hyperbola centered at 0 (Proposition 4.1). Inverting a rectangular hyperbola in a circle around its center gives a lemniscate of Bernoulli, and this is where the lemniscate comes from. Among the terms with $n \geq 3$, only t_4 has a trace that a Möbius transformation maps into a conic (Proposition 4.2).

In Section 5 I look at the pairs $t_m + t_{2m}$ and the zeros of $\zeta(s)$, and Section 6 lists the corrections to the formulas of [Ha]. None of them changes the main observation, which I find a really nice picture of a small symmetry of ζ.

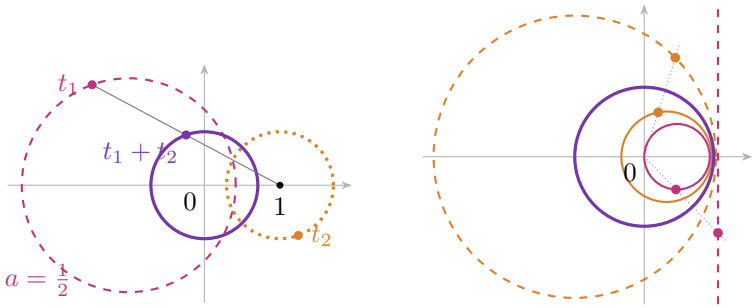


FIGURE 1. Left: on the critical line, t_1 and t_2 trace the dashed and the dotted circle, and $t_1 + t_2$, the midpoint between 1 and t_1, traces the purple circle around 0. Right: the traces of $t_1 + t_2$ for $a = \frac{1}{2}$ (purple), for $a = 0$ and $a = 1$ (pink, solid and dashed) and for $a = \frac{1}{4}$ and $a = \frac{3}{4}$ (orange, solid and dashed). Inversion in the purple circle exchanges the traces for a and $1 - a$, and the dotted rays join mirror points with the same b.

2. The mirror

In the following, I write $z = 2^{-ib} = e^{-ib\log 2}$. As b runs through $\mathbb{R}$, the number z runs through the unit circle, periodically with period $2\pi/\log 2$. Since $2^{1-s} = 2^{1-a}z$ and $2^{-s} = 2^{-a}z$, the first two terms are

$$t_1(s) = \frac{1}{1-2^{1-a}z}\,, \qquad t_2(s) = -\frac{2^{-a}z}{1-2^{1-a}z} = \frac{1-t_1(s)}{2}\,. \tag{2}$$

Notice that $t_1 + t_2 = (1 + t_1)/2$, i.e. the sum is the midpoint between 1 and t_1, as in Figure 1.

Proof of Theorem 🍊. The map $z \mapsto 1/(1-2^{1-a}z)$ is a Möbius transformation. For $a \neq 1$ its pole 2^{a-1} is not on the unit circle, so it maps the unit circle to a circle (see [Ne, Chapter 3]). This circle is symmetric about the real axis, since $t_1(a-ib) = \overline{t_1(a+ib)}$. Therefore its two points $1/(1-2^{1-a})$ and $1/(1+2^{1-a})$ on the real axis, which come from $z = 1$ and $z = -1$, are the ends of a diameter. Their mean and half their distance show that t_1 traces the circle with center $1/(1-4^{1-a})$ and radius $2^{1-a}/|1-4^{1-a}|$. By (2), the trace of $t_1 + t_2$ is the image of this circle under $w \mapsto (1+w)/2$. It has the radius $2^{-a}/|1-4^{1-a}|$ and the center

$$\frac{1}{2}\Big(1+\frac{1}{1-4^{1-a}}\Big) = \frac{1-2^{1-2a}}{1-4^{1-a}}\,.$$

The center is 0 exactly when $2^{1-2a} = 1$, i.e. $a = 1/2$, and then the radius is $2^{-1/2}/|1-2| = 1/\sqrt{2}$. □

For $a = 1$, the pole lies on the unit circle, and $t_1 + t_2$ traces the line $\mathrm{Re}(w) = 3/4$.

The following identity explains the critical line.

Proposition 2.1. *For all s with $2^s \neq 1$ and $2^{1-s} \neq 1$, we have*

$$(t_1(s) + t_2(s))\,\overline{(t_1(1-\overline{s}) + t_2(1-\overline{s}))} = \frac{1}{2}\,.$$

Proof. By (1), we have $t_1(s) + t_2(s) = (1 - 2^{-s})/(1 - 2^{1-s})$. Since $\overline{2^u} = 2^{\overline{u}}$, the complex conjugate of this expression at $1 - \overline{s}$ is

$$\frac{1 - 2^{s-1}}{1 - 2^s} = \frac{-2^{s-1}(1 - 2^{1-s})}{-2^s(1 - 2^{-s})} = \frac{1}{2} \cdot \frac{1 - 2^{1-s}}{1 - 2^{-s}}\,,$$

which is $1/2$ divided by $t_1(s) + t_2(s)$. □

For $s = a + ib$ we have $1 - \overline{s} = (1 - a) + ib$. So Proposition 2.1 says that the inversion $w \mapsto 1/(2\overline{w})$ in the circle $|w| = 1/\sqrt{2}$ maps the value of $t_1 + t_2$ at $a + ib$ to its value at $(1 - a) + ib$. In particular, it exchanges the traces for a and $1 - a$, as in the right half of Figure 1. On the critical line $1 - \overline{s} = s$, and the proposition gives $|t_1(s) + t_2(s)|^2 = 1/2$ directly. The trace for $a = 0$ is the circle with center $1/3$ and radius $1/3$. It passes through 0, and its mirror image is the line $\mathrm{Re}(w) = 3/4$ traced for $a = 1$.

3. Terms that do not trace conics

Let $n \geq 3$ and $a \neq 1$. With $z = 2^{-ib}$ as before, (1) becomes

$$t_n(a + ib) = \frac{(-1)^{n-1} n^{-a}\, n^{-ib}}{1 - 2^{1-a} z}\,. \tag{3}$$

The numerator has modulus n^{-a}, and $|1 - 2^{1-a} z|$ lies between $|1 - 2^{1-a}|$ and $1 + 2^{1-a}$. So the trace lies in the annulus of Theorem 🦄.

Proof of Theorem 🦄. First let n not be a power of 2. Then $\lambda = \log n / \log 2$ is irrational, since $\lambda = p/q$ with positive integers p, q would give $n^q = 2^p$. Let u and v be numbers of modulus 1, and write $v = e^{-i\beta}$ with real β. For $b = (\beta + 2\pi k)/\log 2$ with an integer k, we have $z = v$ and $n^{-ib} = e^{-i\beta\lambda} e^{-2\pi i k \lambda}$. Since λ is irrational, the numbers $k\lambda$ are dense modulo 1, and we can choose k such that n^{-ib} is as close to u as we like. By (3), the trace comes arbitrarily close to every number $(-1)^{n-1} n^{-a} u/(1 - 2^{1-a} v)$. These numbers fill the whole annulus, because u rotates them and $|1 - 2^{1-a} v|$ takes every value between $|1 - 2^{1-a}|$ and $1 + 2^{1-a}$. A conic is closed and contains no disk, while the annulus contains one, since $2^{1-a} > 0$. So the trace does not lie on a conic.

Now let $n = 2^k$ with $k \geq 2$. Then $n^{-ib} = z^k$, and $t_n(a + ib) = W(z)$ with the rational function $W(z) = -n^{-a} z^k/(1 - 2^{1-a} z)$. For $|z| = 1$ we have $\overline{z} = 1/z$, and so $\overline{W(z)} = W^*(z)$ with

$$W^*(z) = -\frac{n^{-a} z^{1-k}}{z - 2^{1-a}}\,.$$

Suppose that the trace lies on a conic. In terms of w and $\overline{w}$, a conic has an equation $Aw^2 + Bw\overline{w} + \overline{A}\,\overline{w}^2 + Cw + \overline{C}\,\overline{w} + D = 0$ with complex A, C and real B, D, not all zero. The rational function

$$R(z) = A\,W(z)^2 + B\,W(z)W^*(z) + \overline{A}\,W^*(z)^2 + C\,W(z) + \overline{C}\,W^*(z) + D$$

vanishes on the unit circle and therefore everywhere. Near $z = 0$, the functions W and WW^* are holomorphic, W^* has a pole of order $k - 1$ and W^{*2} has a pole of order $2k - 2 > k - 1$. Comparing the poles at 0 gives first $A = 0$ and then $C = 0$. Thus $B|W(z)|^2 + D = 0$ on the unit circle. Since $|W(z)|^2 = n^{-2a}/(1 + 4^{1-a} - 2^{2-a}\,\mathrm{Re}(z))$ is not constant there, we get $B = D = 0$, a contradiction. □

For $n = 8$ the trace is again a closed curve, as in Figure 2, but it lies on no conic.

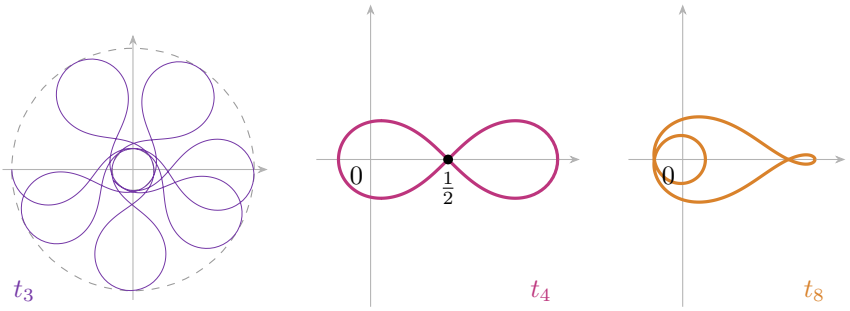


FIGURE 2. The traces on the critical line. Left: t_3 for $0 \le b \le 60$, which fills the dashed annulus densely as b grows. Middle: the lemniscate traced by t_4. Right: the closed curve traced by t_8.

4. The lemniscate

For $n = 4$, (3) reads $t_4(a+ib) = -4^{-a}z^2/(1-2^{1-a}z)$. Throughout this section let $a \neq 1$ and write $\varphi = b\log 2$, so that $z = e^{-i\varphi}$ and $z + 1/z = 2\cos\varphi$.

Proposition 4.1. *If $2\cos(b\log 2) \neq 2^{1-a}$, then $v = 1/(t_4(a+ib) - 4^{-a})$ satisfies*

$$|v + 4^a| = 2^a \left|\mathrm{Re}(v) + 4^a/2\right|.$$

So v lies on the conic with focus -4^a, directrix $\mathrm{Re}(v) = -4^a/2$ and eccentricity 2^a. It is an ellipse for $a < 0$, a parabola for $a = 0$ and a hyperbola for $a > 0$.

Proof. Since $z^2 - 2^{1-a}z + 1 = z\,(2\cos\varphi - 2^{1-a})$, we have

$$t_4(a+ib) - 4^{-a} = -\frac{4^{-a}(z^2 - 2^{1-a}z + 1)}{1 - 2^{1-a}z} = -\frac{4^{-a}z\,(2\cos\varphi - 2^{1-a})}{1 - 2^{1-a}z}.$$

With $1/z = e^{i\varphi}$, this gives

$$v = -\frac{4^a(e^{i\varphi} - 2^{1-a})}{2\cos\varphi - 2^{1-a}}, \qquad v + 4^a = \frac{4^a e^{-i\varphi}}{2\cos\varphi - 2^{1-a}},$$

and also $\mathrm{Re}(v) + 4^a/2 = 2^a/(2\cos\varphi - 2^{1-a})$. Taking absolute values gives the claim. □

Proof of Theorem ♾. Let $a = 1/2$ and write $v = X + iY$. Then Proposition 4.1 says $(X+2)^2 + Y^2 = 2(X+1)^2$, i.e. $X^2 - Y^2 = 2$. So v lies on a rectangular hyperbola centered at 0 (see Figure 3). The two excluded values $\varphi = \pm\pi/4$ modulo 2π give $t_4 = 1/2$, and they cut the period into two arcs. On each arc, v is continuous, and $Y = -2\sin\varphi/(2\cos\varphi - \sqrt{2})$ tends to $+\infty$ at one end and to $-\infty$ at the other. Since $X > 0$ for $|\varphi| < \pi/4$ and $X < 0$ on the other arc, v runs through both branches of the hyperbola. For $p = t_4 - 1/2 = 1/v$ we get

$$\mathrm{Re}(p^2) = \frac{\mathrm{Re}(\overline{v}^2)}{|v|^4} = \frac{X^2 - Y^2}{|v|^4} = 2|p|^4,$$

which is the equation of the theorem. Conversely, every $p \neq 0$ with $\mathrm{Re}(p^2) = 2|p|^4$ is $1/v$ for a point v of the hyperbola, and $p = 0$ comes from $\varphi = \pm\pi/4$. So the trace is the whole lemniscate.

Now let $a \neq 1/2$. For $a = 1$ the trace is unbounded, so we may assume $a \neq 1$. A lemniscate of Bernoulli crosses itself in one point, at a right angle, and removing this point leaves two loops. If the trace were a lemniscate, the continuous map $z \mapsto t_4$ on the unit circle would therefore pass through the crossing point for two different z, since otherwise the rest of the circle would be mapped into one loop. Since $1/t_4 = -4^a(\overline{z}^2 - 2^{1-a}\overline{z})$ for $|z| = 1$, two numbers $z_1 \neq z_2$ of modulus 1 give the same value exactly when $\overline{z}_1 + \overline{z}_2 = 2^{1-a}$. This forces $z_2 = \overline{z}_1$ and $2\,\mathrm{Re}(z_1) = 2^{1-a}$, so a crossing exists only for $a > 0$. It lies at 4^{-a} and comes from $\varphi = \pm\varphi_0$ with $\cos\varphi_0 = 2^{-a}$ and $0 < \varphi_0 < \pi/2$. The function $g(\varphi) = 1/t_4 = -4^a(e^{2i\varphi} - 2^{1-a}e^{i\varphi})$ has the derivative $g'(\pm\varphi_0) = \pm 2 \cdot 4^a \sin(\varphi_0)\, e^{\pm i\varphi_0}$, and $g(\pm\varphi_0) = 4^a$. Since $t_4 = 1/g$, the two branches through 4^{-a} have the directions $e^{i\varphi_0}$ and $e^{-i\varphi_0}$, and they cross at the angle $2\varphi_0$. This is a right angle only for $\varphi_0 = \pi/4$, i.e. $2^{-a} = 1/\sqrt{2}$ and $a = 1/2$. □

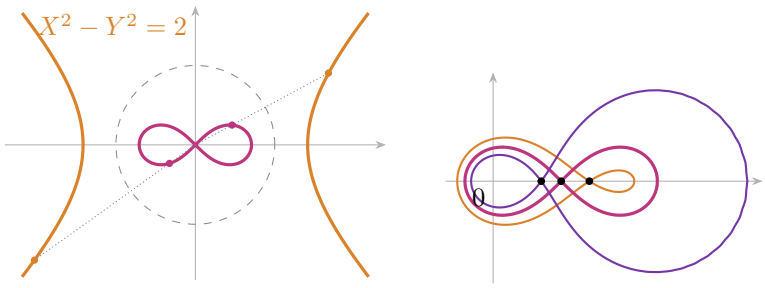


FIGURE 3. Left: inversion in the unit circle maps the hyperbola $X^2 - Y^2 = 2$ to the lemniscate $(x^2 + y^2)^2 = (x^2 - y^2)/2$, and the dotted rays join inverse points. Moved by $1/2$, this lemniscate is the trace of t_4 on the critical line. Right: the traces of t_4 for $a = \frac{1}{4}$ (orange), $a = \frac{1}{2}$ (pink) and $a = \frac{3}{4}$ (purple) cross themselves at 4^{-a}, at the angles 65.5°, 90° and 107.0°.

The next proposition shows that t_4 is the only term with $n \geq 3$ of this kind.

Proposition 4.2. *Let $n \geq 3$ and $a \neq 1$. A Möbius transformation maps the trace of t_n on the line $\mathrm{Re}(s) = a$ into a conic only if $n = 4$.*

Proof. If n is not a power of 2, the closure of the trace contains a disk by Theorem 🦄, and so does the closure of its image under a Möbius transformation, if we take the disk away from the pole. So let $n = 2^k$ with $k \geq 3$, take W and W^* from the proof of Theorem 🦄, a Möbius transformation μ and the one μ^* with conjugate coefficients. Suppose that $F(p, q) = Ap^2 + Bpq + \overline{A}q^2 + Cp + \overline{C}q + D$ vanishes for $p = \mu(W(z))$ and $q = \mu^*(W^*(z)) = \overline{p}$ on $|z| = 1$, and hence for all z. If $\mu(\infty) = \infty$, then μ is affine, and Theorem 🦄 applies to the conic $\mu^{-1}(\{F = 0\})$. Otherwise $q = q_0 + \kappa z^{k-1} + O(z^k)$ near 0 with $\kappa \neq 0$, since W^* has a pole of order $k - 1$.

If $\mu(0) = \infty$, then p has a pole of order k at 0. The term Ap^2 is the only one with a pole of order $2k$, so $A = 0$. What remains is $p\,(Bq + C) + \overline{C}q + D = 0$, and this forces $Bq + C$ to vanish to order k at 0, which gives $B = C = 0$ and then $D = 0$.

If $\mu(0)$ is finite, then $p(z) = p_0 + \lambda z^k + O(z^{k+1})$ with $\lambda \neq 0$. Expanding F around (p_0, q_0), the coefficients of z^0, z^{k-1}, z^k, z^{2k-2} and z^{2k-1} show one after the other that F, $\partial F/\partial q$ and $\partial F/\partial p$ vanish at (p_0, q_0), and that $A = 0$ and $B = 0$. Here we use $k < 2k - 2$, i.e. $k \geq 3$. Then $C = \partial F/\partial p(p_0, q_0) = 0$ and $D = 0$. In all cases $F = 0$, a contradiction. □

For $k = 2$ the powers z^k and z^{2k-2} coincide, and this leaves room for the conic of Proposition 4.1.

5. THE PAIRS AND THE ZEROS

For odd m, (1) gives

$$t_m(s) + t_{2m}(s) = m^{-s}\big(t_1(s) + t_2(s)\big)\,,$$

which is [Ha, (13)]. On the critical line $|m^{-s}| = m^{-1/2}$, and Theorem 🍊 shows that the trace of $t_m + t_{2m}$ is the circle $|w| = 1/\sqrt{2m}$, as in [Ha, Corollary 4.1]. For $m \geq 3$ and $a \neq 1/2, 1$, the argument in the proof of Theorem 🦄 shows that the trace is dense in the set of all $m^{-a}uw$ with $|u| = 1$ and w on the circle of Theorem 🍊. This set is an annulus, or a disk for $a = 0$, and these are the annular rosettes of [Ha].

Notice that the pairs never contain the terms t_n with $4 \mid n$, such as t_4 and t_8. For $\mathrm{Re}(s) > 1$ all series converge absolutely, and $\sum_{m\ \mathrm{odd}} m^{-s} = (1 - 2^{-s})\zeta(s)$ gives

$$\sum_{m\ \mathrm{odd}} \big(t_m(s) + t_{2m}(s)\big) = \frac{(1 - 2^{-s})^2}{1 - 2^{1-s}}\,\zeta(s)\,, \qquad \sum_{k\geq 1} t_{4k}(s) = -\frac{4^{-s}}{1 - 2^{1-s}}\,\zeta(s)\,.$$

Their sum is $\zeta(s)$, since $(1 - 2^{-s})^2 - 4^{-s} = 1 - 2^{1-s}$. On the critical line, the series of pairs does not converge, because $\sum_{m\ \mathrm{odd}} m^{-1/2-ib}$ diverges. The series of all $t_n(s)$ still converges there, but only conditionally, through the cancellation of consecutive terms.

In [Ha], the centered circles are offered as a geometric perspective on the distribution of zeros. However, the circles of single pairs cannot locate zeros. For example, the trace of $t_1 + t_2$ for $a = 0$

passes through 0 (see Figure 1), although its circle is not centered, while the centered circle for $a = 1/2$ never meets 0. Also, the combination

$$(t_1(s) + t_2(s)) - 3^{4/5}(t_3(s) + t_6(s)) = (1 - 3^{4/5-s})(t_1(s) + t_2(s))$$

of two pairs, which both trace centered circles on the critical line, vanishes at $s = 4/5$. More generally, Proposition 2.1 is a property of the single factor $(1-2^{-s})/(1-2^{1-s})$. The map $s \mapsto 1-\overline{s}$ is the reflection in the critical line, which also appears in the functional equation of $\zeta(s)$, but this identity says nothing about the zeros of $\zeta(s)$.

Remark 5.1. The value $1/2$ also depends on the base 2. For an integer $q \geq 2$, the same idea with the factor $1 - q^{1-s}$ gives

$$\zeta(s) = \frac{1}{1-q^{1-s}} \sum_{n=1}^{\infty} \frac{c_q(n)}{n^s}, \qquad c_q(n) = \begin{cases} 1-q & \text{if } q \mid n, \\ 1 & \text{otherwise,} \end{cases}$$

where the series converges for $\mathrm{Re}(s) > 0$. For $q = 2$ this is the series above. If $q \nmid m$, the terms for $n = m$ and $n = qm$ add up to m^{-s} times $(1-(q-1)w)/(1-qw)$ with $w = q^{-s}$. For real $0 < p' < p$, the ratio $|w-p|^2/|w-p'|^2 = (r^2+p^2-2p\,\mathrm{Re}(w))/(r^2+p'^2-2p'\,\mathrm{Re}(w))$ is constant on the circle $|w| = r$ exactly when $r^2 = pp'$, i.e. when p and p' are mirror images in the circle. With the zero $p = 1/(q-1)$ and the pole $p' = 1/q$, the pair has constant modulus on the line $\mathrm{Re}(s) = a$ exactly for $a = \frac{1}{2} + \log(q-1)/(2\log q)$. This is the critical line for $q = 2$, while $q = 3$ and $q = 10$ give $a \approx 0.815$ and $a \approx 0.977$. The mirror of Proposition 2.1 itself works for every q, since $(1-q^{-s})/(1-q^{1-s})$ has modulus $1/\sqrt{q}$ on the critical line.

6. Corrections to the manuscript

I checked the formulas of [Ha] against (1), and I also implemented them in SageMath and checked them numerically with 40 digits at many points. Five points need a correction.

(1) The radius in [Ha, (11)] should be $2^{-a}/|1-4^{1-a}|$ instead of $2^{-a}/|1+2^{1-a}|$. Both agree only for $a = 0$. For $a = 1/2$ the printed formula gives $1 - 1/\sqrt{2} \approx 0.293$ instead of $1/\sqrt{2}$.

(2) In [Ha, Table 1], the radius for $a = 0.75$ is $2^{-1/4} + 2^{-3/4} \approx 1.4355$ instead of $2^{-1/4} + 1/2 \approx 1.341$, which also changes Figure 4 there. For $a = 1.5$ the center is $3/2$ and the radius is $1/\sqrt{2}$, instead of $3/7$ and $2\sqrt{2}/7$.

(3) Formula [Ha, (4)] is not the polar equation of an ellipse with a focus at 0, since its left side is the square of the modulus and $b\log 2$ is not the polar angle of $t_n(a+ib)$. By Theorem 🦄 no trace with $n \geq 3$ lies on a conic, and the annulus in [Ha, Section 2] is the correct picture.

(4) The function $B(z) = (\sqrt{2}-z)/(1-\sqrt{2}z)$ before [Ha, (8)] has modulus 1 on the unit circle, but a pole at $1/\sqrt{2}$ inside the unit disk. It is -1 divided by the Blaschke factor $(z-1/\sqrt{2})/(1-z/\sqrt{2})$.

(5) In [Ha, Section 4], the pairs are said to extend the centered circles to all powers of 2. The pairs only use m and $2m$, and the terms t_n with $4 \mid n$ belong to no pair (see Section 5).

References

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[Ne] T. Needham, *Visual Complex Analysis*, Oxford University Press, 1997.